

```
import 'package:flutter/material.dart';
import 'package:flutter_tts/flutter_tts.dart';
```

```
void main() {
  runApp(MyApp());
}
```

```
class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: "Text To Speech App",
      debugShowCheckedModeBanner: false,
      theme: ThemeData(
        colorScheme: ColorScheme.fromSeed(
          seedColor: Color(0xFF4A90D9),
          brightness: Brightness.light,
        ),
        useMaterial3: true,
      ),
      home: HomePage(),
    );
  }
}
```

```
class HomePage extends StatefulWidget {
  @override
  _HomePageState createState() =>
    _HomePageState();
}
```

```
class _HomePageState extends
State<HomePage> {
  FlutterTts flutterTts = FlutterTts();

  TextEditingController textController =
TextEditingController();

  double speechRate = 0.5;

  int currentIndex = 0;

  bool isSpeaking = false;

  @override
  void initState() {
    super.initState();
    flutterTts.setCompletionHandler(() {
      setState(() => isSpeaking = false);
    });
  }

  void _showErrorDialog(String message) {
    showDialog(
      context: context,
      builder: (_) => AlertDialog(
        shape:
RoundedRectangleBorder(borderRadius:
BorderRadius.circular(18)),
        title: Row(
          children: [
            Icon(Icons.error_outline, color:
Colors.redAccent, size: 28),
            SizedBox(width: 10),
            Text("Invalid Input",
              style: TextStyle(
                color: Colors.redAccent,
                fontWeight: FontWeight.bold)),
          ],
        ),
      ),
    );
  }
}
```

```

    ],
  ),
  content: Text(
    message,
    style: TextStyle(fontSize: 15, color:
Color(0xFF444444)),
  ),
  actions: [
    TextButton(
      onPressed: () =>
Navigator.pop(context),
      child: Text("OK",
        style: TextStyle(
          color: Color(0xFF4A90D9),
fontWeight: FontWeight.bold)),
    ),
  ],
);
}

Future<void> speakText() async {
  final text = textController.text.trim();

  if (text.isEmpty) {
    _showErrorDialog("Please enter some text
before speaking.");
    return;
  }

  // Allow only letters (a-z, A-Z), spaces, and
basic punctuation

```

```

    final validPattern = RegExp(r"^[a-zA-
Z\s,!\?\"-]+\$");
    if (!validPattern.hasMatch(text)) {
      _showErrorDialog(
        "Only letters and basic punctuation are
allowed.\nNumbers and special characters
(e.g. @, #, 1, 2) are not permitted.");
      return;
    }

    await flutterTts.setSpeechRate(speechRate);
    await flutterTts.speak(text);
    setState(() => isSpeaking = true);
  }

Future<void> stopSpeech() async {
  await flutterTts.stop();
  setState(() => isSpeaking = false);
}

String get rateLabel {
  if (speechRate < 0.35) return 'Slow 🐢';
  if (speechRate < 0.7) return 'Normal 🚶';
  return 'Fast 🚀';
}

@override
Widget build(BuildContext context) {
  return Scaffold(
    backgroundColor: Color(0xFFFF0F4F8),
    appBar: AppBar(
      flexibleSpace: Container(

```

```

        decoration: BoxDecoration(
          gradient: LinearGradient(
            colors: [Color(0xFF4A90D9),
Color(0xFF7B5EA7)],
            begin: Alignment.topLeft,
            end: Alignment.bottomRight,
          ),
        ),
      ),
    title: Row(
      children: [
        Icon(Icons.record_voice_over, color:
Colors.white),
        SizedBox(width: 10),
        Text(
          "Text To Speech",
          style: TextStyle(
            color: Colors.white,
            fontWeight: FontWeight.bold,
            fontSize: 20),
        ),
      ],
    ),
    iconTheme: IconThemeData(color:
Colors.white),
  ),
  drawer: Drawer(
    child: ListView(
      padding: EdgeInsets.zero,
      children: [
        DrawerHeader(
          decoration: BoxDecoration(

```

```

          gradient: LinearGradient(
            colors: [Color(0xFF4A90D9),
Color(0xFF7B5EA7)],
            begin: Alignment.topLeft,
            end: Alignment.bottomRight,
          ),
        ),
        child: Column(
          crossAxisAlignment:
CrossAxisAlignment.start,
          mainAxisAlignment:
MainAxisAlignment.end,
          children: [
            CircleAvatar(
              backgroundColor: Colors.white24,
              radius: 28,
              child:
Icon(Icons.record_voice_over,
                color: Colors.white, size: 30),
            ),
            SizedBox(height: 12),
            Text("TTS App",
              style: TextStyle(
                color: Colors.white,
                fontSize: 20,
                fontWeight: FontWeight.bold)),
          ],
        ),
      ),
    ListTile(
      leading:
        Icon(Icons.home, color:
Color(0xFF4A90D9)),

```



```
maxLines: 4,  
decoration: InputDecoration(  
  hintText: "Type something to  
speak...",  
  filled: true,  
  fillColor: Color(0xFFFFF8FF),  
  border: OutlineInputBorder(  
    borderRadius:  
BorderRadius.circular(12),  
    borderSide: BorderSide.none,  
  ),  
  focusedBorder:  
OutlineInputBorder(  
    borderRadius:  
BorderRadius.circular(12),  
    borderSide:  
      BorderSide(color:  
Color(0xFF4A90D9), width: 2),  
  ),  
  suffixIcon: IconButton(  
    icon: Icon(Icons.clear, color:  
Colors.grey),  
    onPressed: () =>  
textController.clear(),  
  ),  
),  
),  
),  
],  
),  
),  
  
        SizedBox(height: 20),
```

```
// — Speech Rate Card —

Container(
  decoration: BoxDecoration(
    color: Colors.white,
    borderRadius:
BorderRadius.circular(18),
    boxShadow: [
      BoxShadow(
        color:
Color(0xFF7B5EA7).withOpacity(0.1),
        blurRadius: 12,
        offset: Offset(0, 4),
      )
    ],
  ),
  padding:
EdgeInsets.symmetric(horizontal: 20, vertical:
16),
  child: Column(
    crossAxisAlignment:
CrossAxisAlignment.start,
    children: [
      Row(
        mainAxisAlignment:
MainAxisAlignment.spaceBetween,
        children: [
          Text("Speech Rate",
            style: TextStyle(
              fontWeight: FontWeight.bold,
              fontSize: 15,
              color: Color(0xFF333333))),
          Container(
            padding:
```



```

    ),
    borderRadius:
BorderRadius.circular(14),
    boxShadow: [
      BoxShadow(
        color:
Color(0xFF4A90D9).withOpacity(0.3),
        blurRadius: 10,
        offset: Offset(0, 4),
      )
    ],
  ),
  child: ElevatedButton.icon(
    style: ElevatedButton.styleFrom(
      backgroundColor:
Colors.transparent,
      shadowColor:
Colors.transparent,
      padding:
EdgeInsets.symmetric(vertical: 16),
      shape:
RoundedRectangleBorder(
        borderRadius:
BorderRadius.circular(14)),
    ),
    icon: Icon(
      isSpeaking ? Icons.volume_up :
Icons.play_arrow,
      color: Colors.white),
    label: Text(
      isSpeaking ? "Speaking..." :
"Speak",
      style: TextStyle(
        color: Colors.white,

```

```

fontWeight: FontWeight.bold,
fontSize: 16),
  ),
  onPressed: isSpeaking ? null :
speakText,
  ),
  ),
  SizedBox(width: 12),
  Container(
    decoration: BoxDecoration(
      color: Colors.white,
      borderRadius:
BorderRadius.circular(14),
      border: Border.all(color:
Colors.redAccent.shade100, width: 1.5),
      boxShadow: [
        BoxShadow(
          color:
Colors.redAccent.withOpacity(0.1),
          blurRadius: 8,
          offset: Offset(0, 3),
        )
      ],
    ),
    child: IconButton(
      icon: Icon(Icons.stop_rounded,
        color: Colors.redAccent, size:
28),
      padding: EdgeInsets.all(14),
      tooltip: "Stop",
      onPressed: stopSpeech,
    ),

```

```

    ),
  ],
),

  SizedBox(height: 16),

  // — Tip —
  Center(
    child: Text(
      "Tip: Swipe the slider to adjust
speed",
      style: TextStyle(color:
Colors.grey.shade500, fontSize: 13),
    ),
  ),
),
),
bottomNavigationBar: NavigationBar(
  selectedIndex: currentIndex,
  onDestinationSelected: (index) =>
    setState(() => currentIndex = index),
  backgroundColor: Colors.white,
  indicatorColor:
Color(0xFF4A90D9).withOpacity(0.15),
  destinations: [
    NavigationDestination(
      icon: Icon(Icons.home_outlined),
      selectedIcon: Icon(Icons.home, color:
Color(0xFF4A90D9)),
      label: "Home",
    ),
    NavigationDestination(
      icon: Icon(Icons.settings_outlined),
      selectedIcon: Icon(Icons.settings, color:
Color(0xFF7B5EA7)),
      label: "Settings",
    ),
  ],
),
);
}
}

```

